

HILBERT SCHEME

TYSON KLINGNER

1. INTRODUCTION

To understand the geometry of a given algebraic variety X we often need to understand some of its subschemes. For example, if $X = \mathbb{P}^n$, then all of its smooth degree d hypersurfaces are diffeomorphic. When studying subschemes one quickly sees that subschemes comes in families, e.g.,

$$V(xy - tz^2) \subseteq \mathbb{P}_{x,y,z}^2 \times \mathbb{A}_t^1$$

defines a family of conics in $\mathbb{P}^2$. Hence, to understand subschemes we will study them in families. We achieve this via the *Hilbert functor*. Once we introduce the Hilbert functor we will show that the Hilbert functor is representable by a projective scheme, which we call the *Hilbert scheme*. In what proceeds we work over $k = \bar{k}$. Every construction extends to the relative setting, but we work over k for simplicity.

2. HILBERT FUNCTOR

Let X/k be a projective variety and let $\mathcal{O}_X(1)$ be a fixed very ample line bundle. Before defining the Hilbert functor we will introduce the *Hilbert polynomial*.

Definition-Proposition 2.1. Let $\mathcal{F}$ be a coherent sheaf on X , then there is a numerical polynomial $P(z) \in \mathbb{Q}[z]$, such that $\chi(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for every $n \in \mathbb{Z}$. We call $P_{\mathcal{F}}$ the *Hilbert polynomial* of $\mathcal{F}$ with respect to $\mathcal{O}_X(1)$. If $Z \subset X$ is a closed subscheme, then we define the *Hilbert polynomial* of Z to be $P_{\mathcal{O}_Z}$.

Proposition 2.2 (Properties of Hilbert Polynomials). *Here we collect some basic properties of Hilbert polynomials*

- (1) *Suppose there is a short exact sequence of coherent sheaves*

$$0 \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow 0.$$

Then,

$$P_{\mathcal{F}}(n) = P_{\mathcal{E}}(n) + P_{\mathcal{G}}(n)$$

- (2) *If $X \rightarrow Y$ is a flat morphism where Y is a projective variety, then each fibre X_y has the same Hilbert polynomial. Moreover, if Y is integral and if the fibres have the same Hilbert polynomial, then $X \rightarrow Y$ is flat.*

By virtue of (2) we can talk about the Hilbert polynomial relative to a projective morphism.

Definition-Proposition 2.3. If $g : Y \rightarrow Z$ is a projective morphism with $\mathcal{O}_{Y/Z}(1)$ relatively ample and Z connected we can define the Hilbert polynomial of $\mathcal{F}$ where $\mathcal{F}$ is a coherent sheaf on Y , which is flat over Z . Indeed, since $\mathcal{F}$ is flat over Z the Hilbert polynomial is constant in the fibres, and in particular

$$Z \ni z \mapsto h^0(Y_z, \mathcal{F}(n)|_{Y_z}) \in \mathbb{N}$$

is constant for $n \gg 0$ so by cohomology and basechange $g_*\mathcal{F}(n)$ is locally free for $n \gg 0$. There is a numerical polynomial $P_{\mathcal{F}}$ such that $\text{rank}(g_*\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for $n \gg 0$. The relative Hilbert polynomial is invariant under pullback:

- (1) If $p : Z' \rightarrow Z$ is any morphism and $\mathcal{F}'$ is the pullback of $\mathcal{F}$ to $Y \times_Z Z'$, then the Hilbert polynomial of $\mathcal{F}$ agrees with the Hilbert polynomial of $\mathcal{F}'$. Hence, we can compute the Hilbert polynomial of $\mathcal{F}$ over any fibre.

Example 2.4. Suppose $X = \mathbb{P}^m$. Then, $\chi(\mathcal{O}_{\mathbb{P}^m}(n)) = \binom{m+n}{m}$ for $n \gg 0$ so the Hilbert polynomial of $\mathbb{P}^m$ is given by

$$P_{\mathbb{P}^m}(n) = \binom{m+n}{m}, \quad n \gg 0$$

In particular,

$$P_{\mathbb{P}^2}(n) = \frac{(n+1)(n+2)}{2}, \quad n \gg 0.$$

If $C \subseteq \mathbb{P}^2$ is a degree d curve, then there is a short exact sequence

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^2}(n-d) \rightarrow \mathcal{O}_{\mathbb{P}^2}(n) \rightarrow \mathcal{O}_C(n) \rightarrow 0$$

and since $P_C(n) = P_{\mathbb{P}^2}(n) - P_{\mathbb{P}^2}(n-d)$, it follows that

$$P_C(n) = dn - \frac{1}{2}d(d-3), \quad n \gg 0.$$

Definition 2.5. Let X/k and $\mathcal{O}_X(1)$ be as above. Then, the *Hilbert functor* is defined by

$$\text{Hilb}(X) : \text{Sch}/k \rightarrow \text{Sets}$$

$$\text{Hilb}(X)(Z) = \left\{ \begin{array}{l} \text{Subschemes } V \subseteq X \times Z \\ \text{which are proper and flat over } Z \end{array} \right\}.$$

If $P \in \mathbb{Q}[z]$ is a numerical polynomial, then we have the subfunctor

$$\text{Hilb}_P(X) : \text{Sch}/k \rightarrow \text{Sets}$$

$$\text{Hilb}_P(X)(Z) = \left\{ \begin{array}{l} \text{Subschemes } V \subseteq X \times Z \\ \text{which are proper and flat over } Z \\ \text{and have Hilbert polynomial } P \end{array} \right\}.$$

Note that

$$\text{Hilb}(X) = \cup_P \text{Hilb}_P(X).$$

3. REPRESENTABILITY

It is a fundamental result in higher dimension geometry that the Hilbert functor is representable.

Theorem 3.1. *Let X/k be a projective variety with very ample line bundle $\mathcal{O}_X(1)$, and let $P \in \mathbb{Q}[z]$ be a numerical polynomial. Then the functor $\text{Hilb}_P(X)$ is represented by a morphism $u : \text{Univ}_P(X) \rightarrow H_P$ where $\text{Univ}_P(X) \subseteq X \times H_P$ and H_P/k is projective. That is,*

$$\text{Hom}(Z, H_P) \ni g \mapsto g^*(u) \in \text{Hilb}_P(X)(Z)$$

defines an isomorphism. We call the representing scheme H_P the Hilbert scheme and by abuse of notation we often write $H_P = \text{Hilb}_P(X)$ understanding we are referencing the scheme.

The proof is deep and relies on several non-trivial result from algebraic geometry. We will only outline the key ideas. Before doing so we will explicitly compute an example.

Example 3.2. We will compute the Hilbert scheme of degree d curves in $\mathbb{P}_k^2$. By a curve we mean a closed subscheme defined by the zero locus of a degree d homogenous polynomial $f \in k[x, y, z]$. Any such curve can be written as $f = a_0x^d + \dots, a_nz^d$ where $n = \binom{d+2}{2} - 1$. By associating $V(f) \mapsto (a_0, \dots, a_n)$ we have a one-to-correspondence

$$\{\text{degree } d \text{ plane curves}\} \simeq \mathbb{P}^n.$$

Recall that any degree d plane curve has Hilbert polynomial

$$P = dn - \frac{1}{2}d(d-3)$$

and we claim that under this correspondence $\text{Hilb}_P(\mathbb{P}^2) = \mathbb{P}^n$. For this to be true we need a universal family. Consider the *tautological family* $\mathcal{C} \subseteq \mathbb{P}^2 \times \mathbb{P}^n$ defined by

$$\mathcal{C} := V(a_0x^d + \dots + a_nz^d)$$

where (x, y, z) are the homogenous coordinates of $\mathbb{P}^2$, and $(a_0, \dots, a_n)$ the homogenous coordinates of $\mathbb{P}^n$. Over every $a = (a_0, \dots, a_n)$ the fibre $\mathcal{C}_a \subseteq \mathbb{P}^2$ is the curve defined by the polynomial with coefficients $a_0, \dots, a_n$. Now, for every k -scheme Z and $V \in \text{Hilb}_P(\mathbb{P}^2)(Z)$ we need unique morphisms filling in the following Cartesian diagram

$$\begin{array}{ccc} V & \xrightarrow{\quad} & \mathcal{C} \\ \downarrow & & \downarrow \\ Z & \xrightarrow{\quad} & \mathbb{P}^n \end{array}$$

We will demonstrate the morphism for when $Z = \text{Spec} A$ is a finite type affine scheme where the general case follows by gluing. Hence, $V \subseteq \mathbb{P}_A^2$ has Hilbert polynomial

$$P_V(n) = dn - \frac{1}{2}d(d-3)$$

and $\mathcal{I}_V$ has Hilbert polynomial

$$P_{\mathcal{I}_V}(n) = \binom{d+2}{2} - P_V(n)$$

Observe that $P_{\mathcal{I}_V}(d) = 1$ so by cohomology and basechange $\pi_*\mathcal{I}_V(d) \rightarrow \text{Spec} A$ is a line bundle. After localizing we may assume that $\pi_*\mathcal{I}_V(d)$ is free with trivialising section

$$f \in H^0(V, \mathcal{I}_V(d)).$$

Since the image f_s of f in over each fibre $s \in \text{Spec} A$ generates the ideal sheaf $\mathcal{I}_{V,s}$ it follows that $\mathcal{I}_V = (f)$ where $f \in A[x, y, z]$. We write $f = a_0x^d + \cdots a_nz^d$ for $a_i \in A$ and let a_i^s denote the image of a_i in $\kappa(s) \simeq k$. Therefore, we obtain a morphism

$$\text{Spec} A \ni s \mapsto (a_0^s, \dots, a_n^s) \in \mathbb{P}^n.$$

From here it is clear we have a Cartesian square

$$\begin{array}{ccc} V & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ Z & \longrightarrow & \mathbb{P}^n \end{array}$$

Therefore, $\text{Hilb}_P(\mathbb{P}^2) = \mathbb{P}^n$ and $\text{Univ}_P(\mathbb{P}^2) = \mathcal{C}$.

4. PROPERTIES OF THE HILBERT SCHEME

Theorem 4.1. *Let X/k be a projective variety with very ample line bundle $\mathcal{O}_X(1)$, and Let $\text{Hilb}_P(X/S)$ denote the Hilbert scheme with fixed Hilbert polynomial P . The Hilbert scheme has the following properties:*

- (1) *The Zariski tangent space to $\text{Hilb}_P(X)$ at $[V] \in \text{Hilb}(X/k)$ is given by $H^0(V, \mathcal{N})$ where $\mathcal{N}$ is the normal sheaf of V in X ;*
- (2) *If V is a locally complete intersection, and if $H^1(V, \mathcal{N}) = 0$, then $\text{Hilb}_P(X)$ is nonsingular at the point $[V]$ with*

$$\dim_{[V]} \text{Hilb}_P(X) = h^0(V, \mathcal{N}).$$

- (3) *In any case, if V is a locally complete intersection, the dimension of H at $[V]$ is at least $h^0(V, \mathcal{N}) - h^1(V, \mathcal{N})$, i.e.,*

$$\dim_{[V]} \text{Hilb}_P(X) \geq h^0(V, \mathcal{N}) - h^1(V, \mathcal{N}).$$

The theorem is deep and instead of prving it we will demonstrate it for the Hilbert scheme from Example 3.2.

Example 4.2. Consider the Hilbert scheme $\text{Hilb}_P(\mathbb{P}^2) = \mathbb{P}^n$ parameterizing degree d plane curves from Example 3.2. Let $X \subset \mathbb{P}_k^2$ be a degree d curve so that $\mathcal{I}_X = \mathcal{O}_{\mathbb{P}^2}(-d)$. We already know that $\mathbb{P}^n$ is smooth of dimension n so we will verify the dimension conditions asserted by the Theorem. Every hypersurface of projective space is a complete

intersection, and $\mathcal{N} \simeq \mathcal{O}_X(d)$ defines a line bundle. Moreover, we have the following short exact sequence

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^2} \rightarrow \mathcal{O}_{\mathbb{P}^2}(d) \rightarrow \mathcal{O}_X(d) \rightarrow 0.$$

By taking the long exact sequence in cohomology and using well-known vanishing results we see $h^0(\mathcal{O}_V(d)) = \binom{d+2}{2} - 1 = n$ and $h^1(\mathcal{O}_V(d)) = 0$. Hence, by (2) $\text{Hilb}_P(\mathbb{P}^2)$ is nonsingular at $[X]$ with dimension n , which agrees with our example.